1 Between-epoch carry: arrowhead admissibility and the DD cycle basis

The Network-mode document establishes the arrowhead normal-equation topology and its exact Schur reduction (the per-station Schur reduction section of the Network-mode document), and the worked estimability example establishes the minimum-constraint datum (the two-receiver estimability example). This document covers the step between them: *how information from outside the current epoch enters an arrowhead system.*

The structure of the problem sets the requirements. A network estimator that solves each epoch independently discards what it has learned about the carrier-phase ambiguities, the only constant quantities in the problem. An estimator that carries them without care destroys either the sparsity that keeps the solve affordable or the datum invariance that gives the product its meaning. Section 1.1 lists what is carried at all and the route each family takes. The design of the fold then has three parts, developed in order:

1. An arrowhead system admits exactly *three* kinds of information fold. The columns the new information touches determine which one applies; its physical meaning does not (Section 1.2).
2. The carried quantity must therefore be gauge-free. The natural gauge-free basis of the ambiguity block is the *cycle space* of the receiver–satellite observation graph: the double differences indexed by the spanning tree's *chords* (Section 1.3).
3. Carrying that basis costs one bounded Woodbury correction per connected component, with closed forms for the posterior and for an exact innovation test (Section 1.4).

Section 1.5 then carries a two-receiver, two-satellite network, the smallest network whose observation graph has a cycle, through every step with exact arithmetic.

The reference configurations (`config/golden/golden_field_refound.json` and `config/golden/golden_sim_refound.json`) do not arm the invariant DD carry developed here (`network.carry_invariant` is false in both); they carry the float ambiguity vehicle instead (the float-ambiguity-vehicle section). The door analysis of Section 1.2 still governs where that vehicle's folds enter. The shared-border carry takes door 1, and the vehicle's per-station block, with its regression on the shared border, enters before the reduction as whitened rows of that station's own design (`CouplingRowBuilder::buildLocalBlock`, `network/carry/Coupling`), so it touches only that station's blocks and is bound by door 2's ordering constraint. No fold of the reference configurations takes door 3.

Remark 1 (Theoretical footing). Time-linking a *selected* set of linear functions $\boldsymbol{z} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{x}$ while re-estimating the remainder fresh each epoch is the generalised Kalman filter of Teunissen, Khodabandeh & Psychas (*J. Geod.* 95:108, 2021) and the generalised least-squares filter of Hou & Zhang (*J. Geod.* 99:3, 2025). Both leave open the construction of the transition $\boldsymbol{\Phi}^{z}$ of the linked functions across a *changing* linked set. Choosing $\boldsymbol{z}$ as the cycle basis closes that gap: every basis change the graph can force is then a unimodular integer matrix (Section 1.3). The spanning-tree S-basis itself is due to Jonge [1], who selects it once and never re-selects it. The construction here extends it to a basis that changes from epoch to epoch.

1.1 The carried families and how they are folded

Two places hold information across epochs: the shared border and each station's local columns. Both fold before the solve of the next epoch and neither adds a station–station coupling.

The shared border. After every kept epoch the posterior value of each border key (satellite clock, satellite phase bias, satellite code bias) and the covariance of the border are stashed (`networkStashCarry`, `network/state/BorderStash`). At the next epoch the fold (`networkInjectD2CarryPrior`, `network/state/BorderPrior`) applies the time update in information form, as a border prior of door 1. The diagonal route gives each key the variance $\mathrm{var} + q\,\Delta t$, with q the rate of the registry kind (`network/assembly/ParamRegistry`). The dense route (`network.shared_carry_full_cov`, on in the reference configurations) folds one information block

$$\boldsymbol{\Lambda} \;=\; (\boldsymbol{P}_{\mathrm{prev}} + \boldsymbol{Q})^{-1}\,, \tag{1}$$

the information-form Kalman time update on the border, where $\boldsymbol{Q}$ is diagonal and $\boldsymbol{P}_{\mathrm{prev}}$ is the stashed covariance in the datum frame of the Schur-reduction section. The dense block keeps the cross-covariances between clock and phase bias that the diagonal route discards. A key absent from the epoch's layout is *marginalised* (its rows and columns of $\boldsymbol{P}_{\mathrm{prev}}$ are not selected) and a key with no matrix row folds on the diagonal. The fold of each key is decided once per epoch (`networkCarryDecide`, `network/state/CarryDisposition`) and read by the border fold and the coupling rows alike. The decision is one of folded, anchored or loose (the temporal clock anchor), reinitialised (the clock monitor's flag), seeded (a parked entry whose column returned inside `network.readmit_seed_horizon_s`, folded at q over the gap since its park), parked, re-referenced (a phase-bias frame the datum moved), retired (the datum of its constellation changed), screened (a non-positive variance), carried by the two-state clock oscillator model (`network.clock_model`, off) or folded by another prior (the spherical-harmonic one, off).

The station-local columns. A local column carried as an independent diagonal prior would count the information it shares with the border twice. The posterior of the joint $(\boldsymbol{x}_s, r)$ factorises into the marginal of $\boldsymbol{x}_s$, which the dense border block carries, and the distribution of r conditional on $\boldsymbol{x}_s$; only the conditional is carried for r. After the solve, the harvest (`networkHarvestLocalCarries`, `network/carry/LocalCarryHarvest`) reads for each carried column c of a station: its value $\hat{r}_c$; its border-known conditional variance $v_c = (\boldsymbol{N}_{rr}^{-1})_{cc}$, which does not depend on the datum (the Schur-reduction section); and its regression on the border, $\boldsymbol{m}_c = \boldsymbol{N}_{sr}\,\boldsymbol{N}_{rr}^{-1}\boldsymbol{e}_c$, restricted to the free (non-datum) shared columns and kept only where $|m_{cj}|$ exceeds a relative threshold (10^{-8}, `kCouplingSparsifyRelThresh`). Both come from one local solve with the cached factor of the station's block. The next epoch carries c as one pseudo-observation row of that station's own design,

$$r_c + \sum_j m_{cj}\,s_j \;=\; \hat{r}_c + \sum_{j\in\mathrm{present}} m_{cj}\,\hat{s}_j, \qquad \sigma^2 \;=\; v_c + q\,\Delta t + \sum_{j\in\mathrm{departed}} m_{cj}^2\,P_{jj}, \tag{2}$$

written in the correction coordinates of the rows (coefficient $+1$ on the local column, m_{cj} on the border). A border term whose column left the layout enters through its marginal variance, the third term of σ^2. The variance is floored (`network.phase_leveling_var_floor_m2` for the ambiguities) and scaled by the carry variance factor of the variance-component monitor. The row is built in PASS B by `CouplingRowBuilder` (`network/carry/Coupling`), before the reduction, so it is a door 2 fold, and combined with the dense border block it reproduces the batch solve for a constant parameter when the carried columns are conditionally independent given the border, that is when $\boldsymbol{N}_{rr}$ is

Table 1: Carried station-local families. The kind is the registry kind (`network/assembly/ParamRegistry`); the file holds the family's PASS B step and its harvest.

family (file)	process noise	gap horizon	route
ambiguity (`LevelingCarry`)	random walk, `network.phase_leveling_q_amb` (default 0, a random constant)	`network.phase_leveling_max_gap_s` (default 120 s, 900 s in the reference configurations)	local block with coupling rows (float vehicle)
receiver code bias (`RxCodeBiasCarry`)	random walk, `network.rx_cb_carry_q` (default $1 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$)	900 s (fixed, `kRxCbMaxGapS`)	local block (`network.carry_rx_code_bias_in_block`); else diagonal station prior
wet delay (`ZwdCarry`)	random walk, `network.zwd_carry_q`	the ambiguity horizon	local block (`network.carry_zwd_in_block`); else conditional row or diagonal prior
slant ionosphere (`SlantIonoCarry`)	random walk per epoch, `network.carry_iono_in_block_q_m2` (default $2.5 \times 10^{-3}\,\mathrm{m^2}$ at L1)	`network.carry_iono_in_block_max_gap_s` (0 takes the ambiguity horizon)	local block
receiver clock and inter-system bias (`RxClockCarry`)	random walk per epoch, `network.carry_rx_clock_q_m2` (default $0.09\,\mathrm{m^2}$), with a drift state for the clock	`network.carry_rx_clock_max_gap_s`	local block
phase colour (`PhaseColourCarry`)	Gauss–Markov, stationary variance σ^2_{zen} at the zenith with an elevation law, τ 300 s (`network.phase_colour_*`)	900 s (`network.phase_colour_max_gap_s`)	local block

diagonal (`DenseCarryRecursiveEqualsBatch` and `ConditionalCarryRecursiveEqualsBatch` in `tests/network/test_network_estimator.cpp`; the diagonal carry does not). The time update that shrinks each present column, its gauge projection and the exact update are in the float-ambiguity-vehicle section. The float vehicle folds the carried columns of a station as one jointly whitened block, which keeps the cross-covariances between them. Each family has a PASS B step that installs its rows and a harvest in the same file; Table 1 lists them.

In the reference configurations the ambiguity and the receiver code-bias families are carried; the wet delay, the slant ionosphere, the receiver clock and the phase colour are re-estimated at every epoch (their settings are off). The stationary standard deviation of a phase-colour state at elevation e is $\sigma_{\mathrm{zen}} / \sin\big(\max(e, e_{\mathrm{floor}})\big)$.

The registry also declares, per family, the events it responds to: a slip, a gap, a datum handover, a decoupling and a re-admission (`ParamEvent`). A carried quantity with a stiff or zero rate needs such a response before it may be armed, because a stiff prior carried past a jump moves the whole solution (Section 1.5); the responses are developed in the event-response and emitted-product section. The remaining carry, the invariant DD carry (`network.carry_invariant`), is the only one that is not a conditional row: it folds a prior on the double-differenced cycle basis after the reduction, door 3, and is developed in Sections 1.3 and 1.4.

1.2 Three doors into an arrowhead system

The epoch's system has the structure of the Schur reduction section: a shared satellite border $\boldsymbol{x}_s$, one private block $\boldsymbol{x}_r$ per station, and zero station–station coupling. Consider a Gaussian prior $\boldsymbol{z} \sim N(\hat{\boldsymbol{z}}, \boldsymbol{P}^z)$ on a linear function $\boldsymbol{z} = \boldsymbol{C}\boldsymbol{x}$ of the epoch's parameters. In information form it is the increment $\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{P}^z)^{-1}\boldsymbol{C}$ to the normal matrix and $\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{P}^z)^{-1}\hat{\boldsymbol{z}}$ to the right-hand side. The block in

which that increment lands determines how it can be folded.

Door 1: the shared border. If $\boldsymbol{C}$ touches only shared columns, the increment lands entirely in $\boldsymbol{N}_{ss}$. Such a fold is *reduction-preserving*:

> **Lemma.** For any symmetric $\Delta\boldsymbol{N}$ and any $\Delta\boldsymbol{u}$ on the shared border,
>
> $$\widetilde{(\boldsymbol{N}_{ss} + \Delta\boldsymbol{N})} = \tilde{\boldsymbol{N}}_{ss} + \Delta\boldsymbol{N}, \qquad \widetilde{(\boldsymbol{u}_s + \Delta\boldsymbol{u})} = \tilde{\boldsymbol{u}}_s + \Delta\boldsymbol{u}.$$
>
> *Proof.* The Schur subtrahend $\sum_r \boldsymbol{N}_{sr}^r (\boldsymbol{N}_{rr}^r)^{-1} \boldsymbol{N}_{sr}^{r\mathrm{T}}$ is a function of $\boldsymbol{N}_{sr}$ and $\boldsymbol{N}_{rr}$ only; $\boldsymbol{N}_{ss}$ enters the reduced-normals equation additively. □

A border prior may therefore be folded *before or after* the reduction with identical results. The between-epoch carry of the satellite clock and biases costs no second Schur pass even when it is applied late, for example between the null-space rank gate and the solve. `addSharedInfo` (`network/solve/NormalEquations`) uses this: when a reduction is already cached, it folds the delta in place and leaves the cache current. The dense form of the same carry (`addSharedPriorDense`, `network/solve/Estimator`) takes a *covariance* rather than an information matrix. A carried key absent from the current epoch's layout must be *marginalised*, a row and column deletion on $\boldsymbol{P}^z$; deleting the same rows of $(\boldsymbol{P}^z)^{-1}$ would instead *condition* on the departed key and bias every surviving parameter.

Door 2: one station's own columns, before the reduction. If $\boldsymbol{C}$ touches only station r's local columns, the increment lands in $\boldsymbol{N}_{rr}^r$. For a diagonal prior on columns $c \in \mathcal{C}$ with information weights $\Lambda_c \geq 0$ and information right-hand sides η_c,

$$\boldsymbol{N}_{rr}^r \mathrel{+}= \sum_{c\in\mathcal{C}} \Lambda_c \, \boldsymbol{e}_c \boldsymbol{e}_c^{\mathrm{T}}, \qquad \boldsymbol{u}_r^r \mathrel{+}= \sum_{c\in\mathcal{C}} \eta_c \, \boldsymbol{e}_c, \tag{3}$$

which the implementation applies as a sequence of rank-one updates of the *cached* $\mathrm{LDL}^{\mathrm{T}}$ factor (`addStationPrior`). The raw $\boldsymbol{N}_{rr}$ is never stored or re-factorised, and a positive Λ_c keeps the factor positive-definite by construction (a zero weight skips the factor update and still adds its right-hand-side increment). Unlike door 1, this fold is *not* reduction-preserving: it changes $\boldsymbol{N}_{rr}$, hence the Schur subtrahend, hence $\tilde{\boldsymbol{N}}_{ss}$. It must be applied *before* the reduction; there is no post-reduce fast path. It is cheap for two reasons: it is orthogonal to every other station (the arrowhead has no station–station block to disturb), and it conditions only $\boldsymbol{x}_r$.

The two doors have opposite ordering constraints: door 1 may be late, door 2 must be early. The asymmetry follows directly from the reduced-normals equation.

Door 3: local columns across several stations. If $\boldsymbol{C}$ touches local columns of two or more stations, the information increment $\boldsymbol{C}^{\mathrm{T}}(\boldsymbol{P}^z)^{-1}\boldsymbol{C}$ has a nonzero (r, r') block. It is representable neither as a border increment nor as a per-station one. Folding it into the unreduced system *destroys the arrowhead*: it fills in the station–station zeros that make the marginalisation exact and cheap.

This is the case that matters in practice. The only ambiguity combinations worth carrying are between-station ones, because a single station's ambiguity is not estimable on its own (the phase part of the two-receiver example). Such a prior is folded *after* the reduction, by Woodbury through the block-diagonal local normals, using the per-station factors the reduction has already computed and left intact. Section 1.4 gives the algebra; the rest of this subsection records the two hazards it creates.

Remark 2 (Post-reduce folds do not compose). Two door-3 folds applied to the same reduced system are each derived against the *pristine* $\boldsymbol{N}_{rr}$. Composing them over coupled local columns is therefore inexact: neither fold sees the other's augmentation of $\boldsymbol{N}_{rr}$, so the composition drops the cross term $\boldsymbol{C}_L \boldsymbol{N}_{rr}^{-1} \boldsymbol{C}_{L,\text{other}}^{\mathrm{T}}$ of the joint middle inverse without any signal. The implementation enforces the exclusion: the sum-to-zero ensemble phase gauge (`foldReducedSumZeroGauges`) and the carried DD priors (`foldReducedDdPriors`) are mutually exclusive, and `NetworkEstimator::solve` throws if both are set. No configuration the loader accepts arms the ensemble gauge any more; the kernel keeps the fold and the exclusion is tested (`DdPriorAndSumZeroGaugeAreMutuallyExclusive`). Under the invariant carry the phase datum is realised through door 2, by the pre-reduce tree pins. These compose *exactly* with a post-reduce door 3 fold, because the fold then reads an already-pinned factor.

The same truncation applies between DD blocks. `foldReducedDdPriors` folds one block per connected component of one (constellation, signal) group, each against the unmodified per-station factors, so the result is exact only when no station is touched by two blocks. A station tracking two signals sits in two blocks, and the fold then makes the block-Jacobi approximation of dropping the cross terms between them.

Remark 3 (A door-3 fold cannot repair a rank deficiency). Let $\delta\boldsymbol{x}_s$ span a null direction of $\tilde{\boldsymbol{N}}_{ss}$, with the matching local response $\delta\boldsymbol{x}_r = -(\boldsymbol{N}_{rr}^{r})^{-1} \boldsymbol{N}_{sr}^{r\mathrm{T}} \delta\boldsymbol{x}_s$. If the carried function is estimable, $\boldsymbol{z} = \boldsymbol{C}_L \boldsymbol{x}_L$ is invariant along that direction, so $\boldsymbol{C}_L\, \delta\boldsymbol{x}_L = -\boldsymbol{G}^{\mathrm{T}} \delta\boldsymbol{x}_s = 0$ with $\boldsymbol{G}$ as defined in Equation (6). Every column of $\boldsymbol{G}$ is therefore orthogonal to the null space, the rank-K update $\boldsymbol{G}\boldsymbol{M}\boldsymbol{G}^{\mathrm{T}}$ adds nothing along it, and the declared S-basis dimension the rank gate tests is unchanged by the carry. A carry that changed it would be carrying a gauge rather than information.

1.3 The observation graph and its cycle space

Door 3 admits only gauge-free functions, and the carried basis is chosen accordingly. The graph structure determines that choice; no numerical criterion enters.

The graph. For a constellation g and a signal s, let $G_{g,s} = (V, \mathcal{E})$ be the graph whose vertices $V = \mathcal{R} \cup \mathcal{S}_g$ are the receivers and the satellites, and whose *arcs* are the receiver–satellite pairs carrying an ambiguity this epoch: $(r, i) \in \mathcal{E}$ iff station r tracks satellite i on signal s. There is at most one arc per pair, so an unordered vertex pair is a unique arc identity (`ArcKey`, `network/ar/ArcGraph`). The graph is *bipartite* by construction: no arc joins two receivers or two satellites.

Trees and chords. A connected component with n_c receivers, m_c satellites and $|\mathcal{E}_c|$ arcs has a spanning tree of $n_c + m_c - 1$ edges. The remaining

$$K_c \;=\; |\mathcal{E}_c| \;-\; (n_c + m_c - 1) \tag{4}$$

arcs are the *chords*. Adding a chord to the tree closes exactly one cycle, its *fundamental cycle*: the chord plus the unique tree path joining its endpoints. The K_c fundamental cycles are a basis of the component's cycle space. Equation (4) is the first Betti number of the component. It is the exact number of independent double differences the component supports, and it is a property of the *graph*, independent of the estimator.

The carried functions. The graph is bipartite, so every cycle alternates receiver–satellite–receiver–satellite, has even length, and admits an alternating signing of its arcs. For chord c with

fundamental cycle $\mathcal{C}_c$, define

$$z_c \;=\; \sum_{(r,i)\,\in\,\mathcal{C}_c} \sigma_{(r,i)}\, N^i_{r,s}, \qquad \sigma_{(r,i)} = \pm 1 \text{ alternating}, \tag{5}$$

where the sign is $+1$ when the arc is traversed from its lower vertex id to its higher one, and receiver ids lie below satellite ids, so the sign alternates around the cycle (`network/ar/ArcGraph`). For the shortest cycle (length four) z_c is a plain double difference $N^{i_1}_{r_1} - N^{i_1}_{r_2} + N^{i_2}_{r_2} - N^{i_2}_{r_1}$, and for a longer fundamental cycle is a higher-order alternating difference on six or more columns.

> **Proposition (gauge-freeness).** z_c is invariant under both ambiguity gauges of the network model: the receiver shift $N^i_r \mapsto N^i_r - d_r$ for all i, and the satellite shift $N^i_r \mapsto N^i_r - c^i$ for all r.
>
> *Proof.* A cycle visits each of its vertices exactly once, and each visited vertex is incident to exactly *two* arcs of the cycle, which by the alternating signing carry opposite signs. A shift attached to that vertex (d_r at a receiver, c^i at a satellite) therefore contributes $(+1)\delta + (-1)\delta = 0$ to Equation (5). Summing over vertices gives invariance. □

Hence z_c touches no shared column: the receiver and satellite phase biases cancel analytically rather than numerically. Three consequences follow, and together they make the cycle basis the carried object:

- **Estimable in every S-system.** z_c is invariant under the datum, so it may be compared across epochs whose datum differs and graded without an S-transform (the minimum-constraint datum section).
- **Integer, and constant between slips.** z_c is an integer combination of integer ambiguities, and its process noise is $q_z = 0$ in the absence of a cycle slip. It is the *only* part of the ambiguity block for which a zero-q time link is physically correct.
- **Basis changes are unimodular.** Two spanning trees of the same arc set give cycle bases related by an integer matrix $\boldsymbol{T}$ with $\det \boldsymbol{T} = \pm 1$, so $\boldsymbol{z'} = \boldsymbol{T}\boldsymbol{z}$, $\boldsymbol{P}^{z\prime} = \boldsymbol{T}\boldsymbol{P}^z\boldsymbol{T}^{\mathrm{T}}$ is an *exact* change of basis with no information loss. Basis churn defeats any carry of the raw ambiguities, whose individual values depend on the datum; in the cycle basis it has no effect by construction.

Graph events are exact operations. Every way the graph can change between epochs maps to an exact operation on the carried $(\hat{\boldsymbol{z}}, \boldsymbol{P}^z)$. The z-store (`ZComponent`, `network/ar/ZStore`) implements each as a primitive and serves as the test oracle of the driver; the producing code does not build it.

Graph event	Operation on the store
Chord dies (arc lost)	Drop that row and column of $\boldsymbol{P}^z$, pure marginalisation
Chord born (arc closes a new cycle)	Append an entry initialised by tap-off $\hat{z}_{\mathrm{new}} = \boldsymbol{f}^{\mathrm{T}}\hat{\boldsymbol{x}}$ with its conditional variance and cross-covariances
Leaf growth (arc brings in a new vertex)	Extends the tree, closes no cycle, touches no z
Tree edge dies	Pivot a covering chord into the tree ($\boldsymbol{z'} = \boldsymbol{T}\boldsymbol{z}$, $\boldsymbol{T}$ unimodular), then kill the arc as a chord. An uncovered tree edge is a cut edge: its death splits the component, which merely re-partitions existing blocks
Components merge	Union the two blocks block-diagonally (they were independent), promote one bridge arc to the tree, tap off any surplus bridges

The network estimator does not apply these as a sequence of events. The per-epoch driver (`network/carry/InvariantCarryDriver`) receives the current epoch's tree/chord partition from the chain-datum election, takes the trees and fundamental cycles from `network/ar/ArcGraph` (`arcgraph::singleTreeOf`, `arcgraph::fundamentalCycle`) and computes the *composed* transform directly. A new chord's fundamental cycle lies in the previous epoch's cycle space if and only if every arc of that cycle was present last epoch; its carried mean is then $\hat{z}_c^{\text{new}} = \sum_j w_j \, \hat{z}_j^{\text{old}}$, with w_j the incidence of old chord j's arc in the new cycle. Assembling $\boldsymbol{W}$ over the contributing components gives $\hat{\boldsymbol{z}}^{\text{prior}} = \boldsymbol{W}\hat{\boldsymbol{z}}^{\text{old}}$ and $\boldsymbol{P}^{z,\text{prior}} = \boldsymbol{W}\boldsymbol{P}^{z}\boldsymbol{W}^{\mathrm{T}}$. A dead chord's column of $\boldsymbol{W}$ is zero (marginalisation). Merges and splits follow from the contributor gather with no special case. `networkBuildCarryDdPriors` (`network/carry/DdPriorBuild`) resolves each arc to an estimator (station, local ambiguity column), marks the tree edges as the pinned gauge and the others as chords, and excludes GLONASS, whose FDMA signals have no common-wavelength integer double difference. `networkInstallCarry` installs the posterior after the accepted solve, so a withheld or contaminated epoch never enters the carry.

A chord whose cycle uses a newborn arc is *fresh*. The driver gives it a finite diffuse prior (variance `fresh_prior_var`, default 1.0e6 in state units squared) rather than the exact zero-information tap-off of the table. Its posterior is therefore the data-only tap-off up to a term of the order of the ratio of the data variance to `fresh_prior_var`. An arc flagged as slipped this epoch counts as absent from the previous arc set, so every chord whose cycle uses it is also born fresh. The production caller passes only arcs that resolve to an estimator column; a carried chord whose cycle has an unresolvable arc this epoch is dropped from the carry and re-tapped when the arc returns.

Remark 4 (Why the tree must avoid slipped arcs). A slip on a *chord* costs exactly one carried DD. A slip on a *tree edge* corrupts every fundamental cycle routed through it. With `network.phase_datum_eq34_slip_aware_election` armed (default off), the election therefore keeps slipped arcs out of the tree wherever the graph allows it. The argument is structural and independent of any detector threshold; the setting measured net-harmful on a 63-station case and is off (`docs/record/SPEC_d5_churn_limit_cycle.md`).

1.4 Folding the carried double differences

Let one connected component carry K double differences with prior mean $\hat{\boldsymbol{z}}$ and covariance $\boldsymbol{P}^{z}$, expressed as $\boldsymbol{z} = \boldsymbol{C}_L\boldsymbol{x}_L$ where $\boldsymbol{C}_L$ is the $K \times n_L$ signed incidence matrix of Equation (5) over the epoch's local ambiguity columns, and $\boldsymbol{C}_{L,r}$ is its restriction to station r. This is a door-3 fold. Woodbury through the block-diagonal local normals gives, per component,

$$\boldsymbol{B} = \sum_r \boldsymbol{C}_{L,r}\boldsymbol{N}_{rr}^{-1}\boldsymbol{C}_{L,r}^{\mathrm{T}}, \quad \boldsymbol{G} = \sum_r \boldsymbol{N}_{sr}\boldsymbol{N}_{rr}^{-1}\boldsymbol{C}_{L,r}^{\mathrm{T}}, \quad \boldsymbol{z}_{\text{free}} = \sum_r \boldsymbol{C}_{L,r}\boldsymbol{N}_{rr}^{-1}\boldsymbol{u}_r, \quad \boldsymbol{M} = (\boldsymbol{P}^{z} + \boldsymbol{B})^{-1}, \tag{6}$$

and the reduced system takes the rank-K correction

$$\boxed{\begin{aligned} \tilde{\boldsymbol{N}}_{ss} &\mathrel{+}= \boldsymbol{G}\,\boldsymbol{M}\,\boldsymbol{G}^{\mathrm{T}}, && (7)\\ \tilde{\boldsymbol{u}}_s &\mathrel{+}= \boldsymbol{G}\,\boldsymbol{M}\,(\boldsymbol{z}_{\text{free}} - \hat{\boldsymbol{z}}). && (8) \end{aligned}}$$

Every ingredient of Equation (6) comes from the *cached* per-station $\mathrm{LDL}^{\mathrm{T}}$ factors. No re-reduction is needed and the factors are left unmodified. The only new inverse is the $K \times K$ middle matrix $\boldsymbol{M}$, which is per component and bounded in size. Back-substitution then returns the *constrained* locals,

$$\boldsymbol{x}_r \mathrel{-}= \boldsymbol{N}_{rr}^{-1}\boldsymbol{C}_{L,r}^{\mathrm{T}}\,\boldsymbol{M}\,\big(\boldsymbol{z}_{\text{free}} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{x}_s - \hat{\boldsymbol{z}}\big), \tag{9}$$

so the ambiguities read by the quality-control layer and by the next epoch's tap-off are in the same frame as the fold.

Posterior. The value the store carries forward is the prior shrunk towards the data-implied DD,

$$\hat{\boldsymbol{z}}^{+} = \hat{\boldsymbol{z}} + \boldsymbol{P}^{z}\,\boldsymbol{M}\,\boldsymbol{v}, \qquad \boldsymbol{v} = \boldsymbol{z}_{\text{free}} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{x}_{s} - \hat{\boldsymbol{z}}, \tag{10}$$

with covariance, in the datum-as-known shared frame $\boldsymbol{\Sigma}_{ss}$ (the datum frame of the Schur-reduction section: the solve's shared covariance with every datum-pinned and gauge column treated as exact),

$$\boldsymbol{P}^{z+} = \underbrace{\left(\boldsymbol{P}^{z} - \boldsymbol{P}^{z}\boldsymbol{M}\boldsymbol{P}^{z}\right)}_{\text{conditional on } \boldsymbol{x}_{s}} + \left(\boldsymbol{P}^{z}\boldsymbol{M}\boldsymbol{G}^{\mathrm{T}}\right)\boldsymbol{\Sigma}_{ss}\left(\boldsymbol{G}\boldsymbol{M}\boldsymbol{P}^{z}\right). \tag{11}$$

The first term is the posterior given the border; the second restores the coupling through it. $\boldsymbol{\Sigma}_{ss}$ is datum-constrained, but $\boldsymbol{P}^{z+}$ is datum-invariant because $\boldsymbol{z}$ is estimable. `ddPriorPosteriorMean` and `ddPriorPosteriorCovariance` evaluate the two expressions per block from the cached fold factors. The driver builds the datum frame once per install (`NetworkEstimator::datumFrameSharedCov`) and passes it to every block.

Innovation. The same factors give an exact test of the carry against the epoch's data (`ddPriorInnovation`). Under the null hypothesis that the carried DD is still valid,

$$\boldsymbol{v} = \boldsymbol{z}_{\text{free}} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{x}_{s} - \hat{\boldsymbol{z}}, \qquad \mathrm{Cov}(\boldsymbol{v}) = \boldsymbol{M}^{-1} - \boldsymbol{G}^{\mathrm{T}}\,\boldsymbol{\Sigma}_{ss}\,\boldsymbol{G}, \tag{12}$$

and $|v_i|/\sqrt{\mathrm{Cov}(\boldsymbol{v})_{ii}}$ is the standardised statistic that flags a slipped carried DD for re-tapping. With `network.carry_invariant_innov_gate` armed (default off), the driver flags a chord when the statistic exceeds `network.carry_invariant_innov_k` (default 4.0) times a scale factor: 1 by default, or the LS-VCE estimate of the empirical innovation scale when `network.carry_invariant_innov_calibrate` is armed. The *minus* sign in Equation (12) is exact rather than a conservative approximation. $\boldsymbol{x}_s$ is itself influenced by the prior through Equation (8), and subtracting $\boldsymbol{G}^{\mathrm{T}}\boldsymbol{\Sigma}_{ss}\boldsymbol{G}$ corrects for testing the carry against a solution the carry helped produce. Consistency with Equations (10) and (11) is direct: $\boldsymbol{P}^{z}\boldsymbol{M}\,\mathrm{Cov}(\boldsymbol{v})\,\boldsymbol{M}\boldsymbol{P}^{z} = \boldsymbol{P}^{z} - \boldsymbol{P}^{z+}$.

The hard-constraint limit. Setting $K = 1$, $\hat{z} = 0$ and $\boldsymbol{P}^{z} \to \boldsymbol{0}$ turns Equations (7) and (8) into the rank-one fold of an exact linear constraint $\sum_r z_r = 0$ on local columns, with $\boldsymbol{M} \to \boldsymbol{B}^{-1}$. That is the anchor-free sum-to-zero phase-bias gauge (`foldReducedSumZeroGauges`). The DD prior is its soft generalisation: the two are the same object, which is why they cannot be composed.

Positive-definiteness, and why it is counted. $\boldsymbol{P}^{z} + \boldsymbol{B}$ is positive-definite in exact arithmetic, but it can fail to be numerically. A stiff carry ($\boldsymbol{P}^{z}$ near zero) added to an ill-conditioned $\boldsymbol{B}$ subtracts operands of similar magnitude to reach a much smaller result. When the screen fails, the fold is applied *inertly*, $\boldsymbol{M} := \boldsymbol{0}$: the reduced system is untouched and the posterior equals the prior. The block keeps its slot, so the per-block readouts stay index-aligned with the carry's block list.

The inert branch raises no error, and that makes it hazardous. The component's carried sub-state stops taking data, while every downstream reading of it (its formal σ, the innovation test, any ambiguity-resolution probe) continues to report on a quantity that is no longer updated. A carry can therefore diffuse for hundreds of epochs behind clean logs. The implementation treats the PD screen as *always-on telemetry*: per-fold counts of degenerate blocks and their dimension, the worst relative pivot, and the count of repairs are exposed unconditionally (`DdFoldStats`, `network/`

`solve/NormalEquations`). An optional PD-safe path (`network.carry_invariant_pd_safe`, default off) replaces the absolute screen with an equilibrated factorisation and a scale-relative pivot test, evaluates the posterior in a product form that avoids the subtraction, and applies a counted eigenvalue-clamp repair to round-off-level failures. The project rule is that run-end counters are the telemetry of an armed path (`CLAUDE.md`, the pre-flight invariants for estimation changes): an inert path needs a counter, because the absence of an error is no evidence that the mechanism ran.

1.5 Worked example: the smallest network with a cycle

This subsection carries a complete network through every step above with exact arithmetic. The network is the smallest one whose observation graph has a cycle: two receivers, two satellites, one signal. As in the two-receiver estimability example, everything inessential is removed: known coordinates (so the geometric range drops out), one frequency, no atmosphere, noise-free data and unit weights, so every number below is exact. Ambiguities are in cycles.

Model and truth. With geometry and the receiver/satellite clocks removed by the code observations, the carrier phase reduces to

$$\Phi_r^i \;=\; \beta_r \;+\; b^i \;+\; N_r^i, \qquad r \in \{A, B\}, \quad i \in \{1, 2\}, \tag{13}$$

where β_r is the receiver phase bias, b^i the satellite phase bias (the quantity the network emits), and $N_r^i \in \mathbb{Z}$ the ambiguity. The truth is

$$\beta_A = 0.30, \quad \beta_B = -0.20, \qquad b^1 = 0.10, \quad b^2 = -0.15,$$

$$N_A^1 = 3, \quad N_A^2 = -1, \quad N_B^1 = 5, \quad N_B^2 = 2,$$

which through Equation (13) gives the four observations

Φ_r^i	$i = 1$	$i = 2$
$r = A$	3.40	−0.85
$r = B$	4.90	1.65

The graph, its tree, and its one chord. The observation graph has $n_c = 2$ receivers, $m_c = 2$ satellites and $|\mathcal{E}_c| = 4$ arcs, so by Equation (4) its spanning tree has $2 + 2 - 1 = 3$ edges and it has exactly $K_c = 1$ chord (Figure 1). With the tree $\{(A, 1), (A, 2), (B, 2)\}$, the chord is $(B, 1)$, and its fundamental cycle $B \to 1 \to A \to 2 \to B$ gives the single carried function

$$z \;=\; N_B^1 - N_A^1 + N_A^2 - N_B^2. \tag{14}$$

Two exact checks of the Proposition follow. From the truth, $z = 5 - 3 + (-1) - 2 = -1$, an integer as required. From the *observations* alone, with nothing estimated,

$$(\Phi_B^1 - \Phi_B^2) - (\Phi_A^1 - \Phi_A^2) = 3.25 - 4.25 = -1,$$

because β_A, β_B, b^1 and b^2 all cancel in the double difference. The carry transports this single number -1; every other parameter in this subsection depends on the datum.

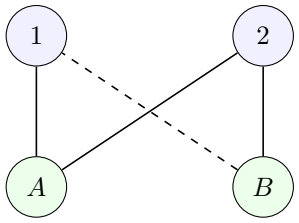


Figure 1: The smallest observation graph with a cycle: two receivers, two satellites, four arcs. Three arcs form a spanning tree; the fourth is a chord, and its fundamental cycle is the one estimable double difference Equation (14).

The tree pins are the datum; the chord is left free. Model Equation (13) has $2+2+4=8$ unknowns and 4 observations, hence a four-dimensional null space (choose $\delta\beta_A, \delta\beta_B, \delta b^1, \delta b^2$ freely; the δN_r^i follow). Pinning the three *tree* ambiguities $N_A^1 = N_A^2 = N_B^2 = 0$ removes three of those four dimensions, leaving the single direction $(\beta_A, \beta_B, b^1, b^2) \mapsto (\beta_A + t,\ \beta_B + t,\ b^1 - t,\ b^2 - t)$, the one remaining receiver-versus-satellite gauge (the network's clock gauge in miniature). One *shared* pin, $b^1 \equiv 0$, removes it.

These four arcs contain the whole datum architecture. The tree edges are pinned through door 2 (locally, before the reduction), the last gauge is fixed by the shared minimum-constraint datum, and the chord is left free: it is the only ambiguity that carries information rather than convention. The pins are taken here as *exact* to keep the arithmetic simple. The implementation realises them as tight finite priors folded by `addStationPrior`, so the pinned columns remain present in $\boldsymbol{N}_{rr}$. The exact-pin idealisation has one visible consequence: with the tree ambiguities identically zero, $\boldsymbol{C}_L$ in Equation (14) collapses onto the chord's own column. With finite pins it retains all four entries, and the fold algebra below has the same form.

Arrowhead blocks. The shared block is ordered $\boldsymbol{x}_s = [\,b^1, b^2\,]^{\mathrm{T}}$ and the free locals $\boldsymbol{x}_A = [\,\beta_A\,]$, $\boldsymbol{x}_B = [\,\beta_B,\ N_B^1\,]^{\mathrm{T}}$. With unit weights the blocks of the per-station block definitions are

$$\boldsymbol{N}_{ss}^{A} = \boldsymbol{I}_2, \quad \boldsymbol{N}_{sr}^{A} = \begin{bmatrix}1\\1\end{bmatrix}, \quad \boldsymbol{N}_{rr}^{A} = \begin{bmatrix}2\end{bmatrix}, \quad \boldsymbol{u}_s^{A} = \begin{bmatrix}3.40\\-0.85\end{bmatrix}, \quad \boldsymbol{u}_r^{A} = \begin{bmatrix}2.55\end{bmatrix},$$

$$\boldsymbol{N}_{ss}^{B} = \boldsymbol{I}_2, \quad \boldsymbol{N}_{sr}^{B} = \begin{bmatrix}1 & 1\\1 & 0\end{bmatrix}, \quad \boldsymbol{N}_{rr}^{B} = \begin{bmatrix}2 & 1\\1 & 1\end{bmatrix}, \quad \boldsymbol{u}_s^{B} = \begin{bmatrix}4.90\\1.65\end{bmatrix}, \quad \boldsymbol{u}_r^{B} = \begin{bmatrix}6.55\\4.90\end{bmatrix},$$

with $(\boldsymbol{N}_{rr}^{B})^{-1} = \left[\begin{smallmatrix}1 & -1\\-1 & 2\end{smallmatrix}\right]$. There is no A–B block: the stations share only b^1 and b^2.

Schur reduction. Eliminating each station's locals through the reduced-normals equation gives

$$\boldsymbol{N}_{ss}^{A} - \boldsymbol{N}_{sr}^{A}(\boldsymbol{N}_{rr}^{A})^{-1}\boldsymbol{N}_{sr}^{A\mathrm{T}} = \begin{bmatrix}0.5 & -0.5\\-0.5 & 0.5\end{bmatrix}, \qquad \boldsymbol{u}_s^{A} - \boldsymbol{N}_{sr}^{A}(\boldsymbol{N}_{rr}^{A})^{-1}\boldsymbol{u}_r^{A} = \begin{bmatrix}2.125\\-2.125\end{bmatrix},$$

while station B contributes exactly zero to both: it has two observations and two free locals, so eliminating the locals absorbs its rows entirely. Summing,

$$\tilde{\boldsymbol{N}}_{ss} = \begin{bmatrix}0.5 & -0.5\\-0.5 & 0.5\end{bmatrix}, \qquad \tilde{\boldsymbol{u}}_s = \begin{bmatrix}2.125\\-2.125\end{bmatrix}, \qquad \mathrm{eig}(\tilde{\boldsymbol{N}}_{ss}) = \{0,\ 1\}. \tag{15}$$

The reduced system has rank 1 and one zero eigenvalue, with eigenvector $[1, 1]^{\mathrm{T}}$: the single gauge predicted above. A null-space rank gate that declares dimension 1 for this configuration passes; one that declares 0 or 2 throws (the minimum-constraint datum section).

Datum, solve, back-substitute. Applying the shared pin $b^1 \equiv 0$ to Equation (15) and solving gives $\hat{b}^2 = -4.25$, and back-substituting through the back-substitution equation,

$$\hat{\beta}_A = 3.40, \qquad \hat{\beta}_B = 5.90, \qquad \hat{N}_B^1 = -1.00.$$

The fit is exact (all four residuals vanish), and the result agrees to machine precision with a dense stacked solve of the same system: the lossless-marginalisation claim of the Schur reduction section in numbers.

These estimates do not resemble the truth. The estimated satellite phase bias $\hat{b}^2 = -4.25$ differs from $b^2 = -0.15$, and $\hat{N}_B^1 = -1$ differs from $N_B^1 = 5$; each has absorbed the datum. The chord's double difference in this frame is

$$\hat{z} = \hat{N}_B^1 - \hat{N}_A^1 + \hat{N}_A^2 - \hat{N}_B^2 = -1 - 0 + 0 - 0 = -1,$$

identical to the truth and to the value read directly off the observations. Datum invariance here is an identity, not an asymptotic property.

Folding a carried DD. Suppose the previous epoch carried this chord. With $\boldsymbol{C}_L = [\,0 \;\; 1\,]$ on station B (one row, the exact-pin collapse noted above), Equation (6) gives

$$\boldsymbol{B} = 2, \qquad \boldsymbol{G} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \boldsymbol{z}_{\text{free}} = 3.25,$$

and the identity $z = \boldsymbol{z}_{\text{free}} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{x}_s = 3.25 - 4.25 = -1$ recovers the DD: $\boldsymbol{z}_{\text{free}}$ is a data-only quantity in the correction frame, not the DD itself. Also $\boldsymbol{G}^{\mathrm{T}}[1,1]^{\mathrm{T}} = 0$: the fold direction is orthogonal to the gauge, as the second Remark of Section 1.2 requires, so no choice of prior can make this carry fix the datum.

A consistent carry is a no-op. With $\hat{z} = -1$ and a rigid prior $\boldsymbol{P}^z = 0$, Equations (7) and (8) reproduce $\hat{b}^2 = -4.25$, $\hat{\beta}_A = 3.40$, $\hat{\beta}_B = 5.90$, $\hat{N}_B^1 = -1$ and zero residuals. The innovation is $v = 3.25 - 4.25 - (-1) = 0$. Carrying information the data already implies changes nothing.

A stale carry distorts the whole solution. Suppose instead that the chord slipped by one cycle undetected, so $\hat{z} = 0$ is folded rigidly. Re-solving gives

	$\hat{b}^2$	$\hat{\beta}_A$	$\hat{\beta}_B$
consistent carry ($\hat{z} = -1$)	-4.25	3.40	5.90
stale carry ($\hat{z} = 0$)	-3.75	3.15	5.15

and post-fit residuals of $+0.25$, -0.25, -0.25, $+0.25$ cycles on rows $(A,1)$, $(A,2)$, $(B,1)$, $(B,2)$ respectively. The one-cycle error in a stiff carried quantity does not stay in the ambiguity that slipped. It moves the *emitted satellite phase bias* by half a cycle and spreads the remainder evenly over every observation in the network. This is the general mechanism by which a stale zero-q carry damages a solution: the network cannot reject information asserted with infinite confidence, so it redistributes the contradiction into whatever states can absorb it. It is the reason for the project rule that any zero-q or stiff time link requires a wired, tested slip, jump or event response before it may be armed (`CLAUDE.md`, the pre-flight invariants for estimation changes). The same rule applies to the edges of the carry: every mechanism that changes which information reaches which states is an event at both edges, with counters and identities. The driver treats a slipped or unresolvable arc as a decoupling event (the chord is dropped or born fresh, and counted), and a returning arc re-couples only through a fresh tap-off.

The innovation test, and its limits. From Equation (12), with $\boldsymbol{\Sigma}_{ss} = \mathrm{diag}(0, 1)$ (the pinned column treated as exact),

$$v = 3.25 - 3.75 - 0 = -0.5, \qquad \mathrm{Cov}(v) = \boldsymbol{M}^{-1} - \boldsymbol{G}^{\mathrm{T}}\boldsymbol{\Sigma}_{ss}\boldsymbol{G} = 2 - 1 = 1, \qquad \frac{|v|}{\sqrt{\mathrm{Cov}(v)}} = 0.5.$$

The test detects the slip at only 0.5σ, far below any usable flag threshold (the default `network.carry_invariant_innov_k` is 4.0). The residuals show why: with four observations and almost no redundancy, the parameters absorbed half the error, so only half of it reached the innovation. The statistic is exact rather than conservative; its *power* is a property of the network's redundancy, and a two-by-two network has almost none. Any estimation-side slip test is diagnostic of the carry only in proportion to the part of the error the network cannot absorb.

A tree-edge death: the unimodular pivot. The two-by-two network is too small to show a pivot: with one chord, every tree edge whose death matters is a cut edge. A third satellite gives $n_c = 2$, $m_c = 3$, $|\mathcal{E}_c| = 6$, so the tree has four edges and there are $K_c = 2$ chords. The tree $\{(A,1),(A,2),(A,3),(B,1)\}$ with chords $(B,2)$ and $(B,3)$ gives

$$z_1 = N_B^2 - N_A^2 + N_A^1 - N_B^1, \qquad z_2 = N_B^3 - N_A^3 + N_A^1 - N_B^1.$$

Let the tree edge $(A,1)$ die. Both chords cover it, so one covering chord, $(B,2)$, is promoted into the tree and the dead arc is killed as a chord. Against the new tree $\{(A,2),(A,3),(B,1),(B,2)\}$ the cycle basis is

$$z_1' = N_A^1 - N_B^1 + N_B^2 - N_A^2 = z_1, \qquad z_2' = N_B^3 - N_A^3 + N_A^2 - N_B^2 = z_2 - z_1,$$

that is $\boldsymbol{z}' = \boldsymbol{T}\boldsymbol{z}$ with

$$\boldsymbol{T} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \qquad \det \boldsymbol{T} = 1.$$

$\boldsymbol{T}$ is integer with unit determinant, so $\boldsymbol{P}^{z\prime} = \boldsymbol{T}\boldsymbol{P}^{z}\boldsymbol{T}^{\mathrm{T}}$ is an exact change of basis: the carried covariance is transformed without approximation. Killing the dead arc's row of $\boldsymbol{z}'$ is then a marginalisation, and the surviving z_2' continues with no loss. A carry of the raw ambiguities would require an S-transformation of a datum-dependent state for the same graph event; the cycle basis removes that requirement.

1.6 Where each object lives

Object	Symbol	Home
Arrowhead accumulation, Schur reduction, back-substitution	`addStation`, `reduce`, `reconstructStation`	`network/solve/NormalEquations`
Door 1: shared border prior	`addSharedInfo`, `addSharedPriorDense`	`network/solve/NormalEquations`, `network/solve/Estimator`
Border carry: stash, decision, fold	`networkStashCarry`, `networkCarryDecide`, `networkInjectD2CarryPrior`	`network/state/BorderStash`, `CarryDisposition`, `BorderPrior`
Door 2: pre-reduce local prior and coupling row	`addStationPrior`, `CouplingRowBuilder`	`network/solve/NormalEquations`, `network/carry/Coupling`
Harvest of the station-local carries	`networkHarvestLocalCarries`, `networkHarvestColumn`	`network/carry/LocalCarryHarvest` and the family files
Door 3: post-reduce Woodbury folds	`foldReducedSumZeroGauges`, `foldReducedDdPriors`	`network/solve/NormalEquations`
Cycle algebra (trees, fundamental cycles, chord restriction)	`ArcKey`, `arcgraph::singleTreeOf`, `arcgraph::fundamentalCycle`	`network/ar/ArcGraph`
Lifecycle primitives (oracle)	`ZComponent`	`network/ar/ZStore`
Per-epoch reconcile $\boldsymbol{W}$, posterior install, innovation gate	`buildDdPriors`, `installPosterior`	`network/carry/InvariantCarryDriver`, `DdPriorBuild`, `InstallCarry`
Posterior and innovation readouts	`ddPriorPosteriorMean`, `ddPriorPosteriorCovariance`, `ddPriorInnovation`	`network/solve/NormalEquations`
PD screen telemetry	`DdFoldStats`, `ddPriorBlockDegenerate`	`network/solve/NormalEquations`
Datum and rank gate	`NetworkSbasis`, `assertNullSpaceRank`	`network/solve/Sbasis`

References

[1] P. J. de Jonge. "A processing strategy for the application of the GPS in networks". PhD thesis. Delft, The Netherlands: Delft University of Technology, 1998.